

Customer Shopping Behavior Analysis

Uncovering insights from 3,900 purchases to drive data-driven business decisions





Project Overview



3,900 Purchases

Transactional data across multiple product categories



Key Objectives

Spending patterns, preferences, subscriptions, trends



Business Impact

Marketing strategies, product positioning, customer retention



Dataset Summary

Dataset Details

- 3,900 total records
- 18 columns of data
- 37 missing values handled

Key Features

- Demographics: Age, Gender, Location
- Purchase details: Category, Amount, Season
- Behavior: Discounts, Reviews, Frequency

Data Preparation Process

01

Data Loading

Imported using Pandas library

02

Exploration

`df.info()` and `df.describe()` analysis

03

Cleaning

Median imputation for missing ratings

04

Standardization

Converted to snake_case format

05

Feature Engineering

Age groups, purchase frequency variables

06

Database Integration

Loaded into PostgreSQL

SQL Analysis Highlights

1

Revenue by Gender

Male: \$157,890 | Female: \$75,191

2

Top Products by Rating

Gloves (3.86), Sandals (3.84), Boots (3.82)

3

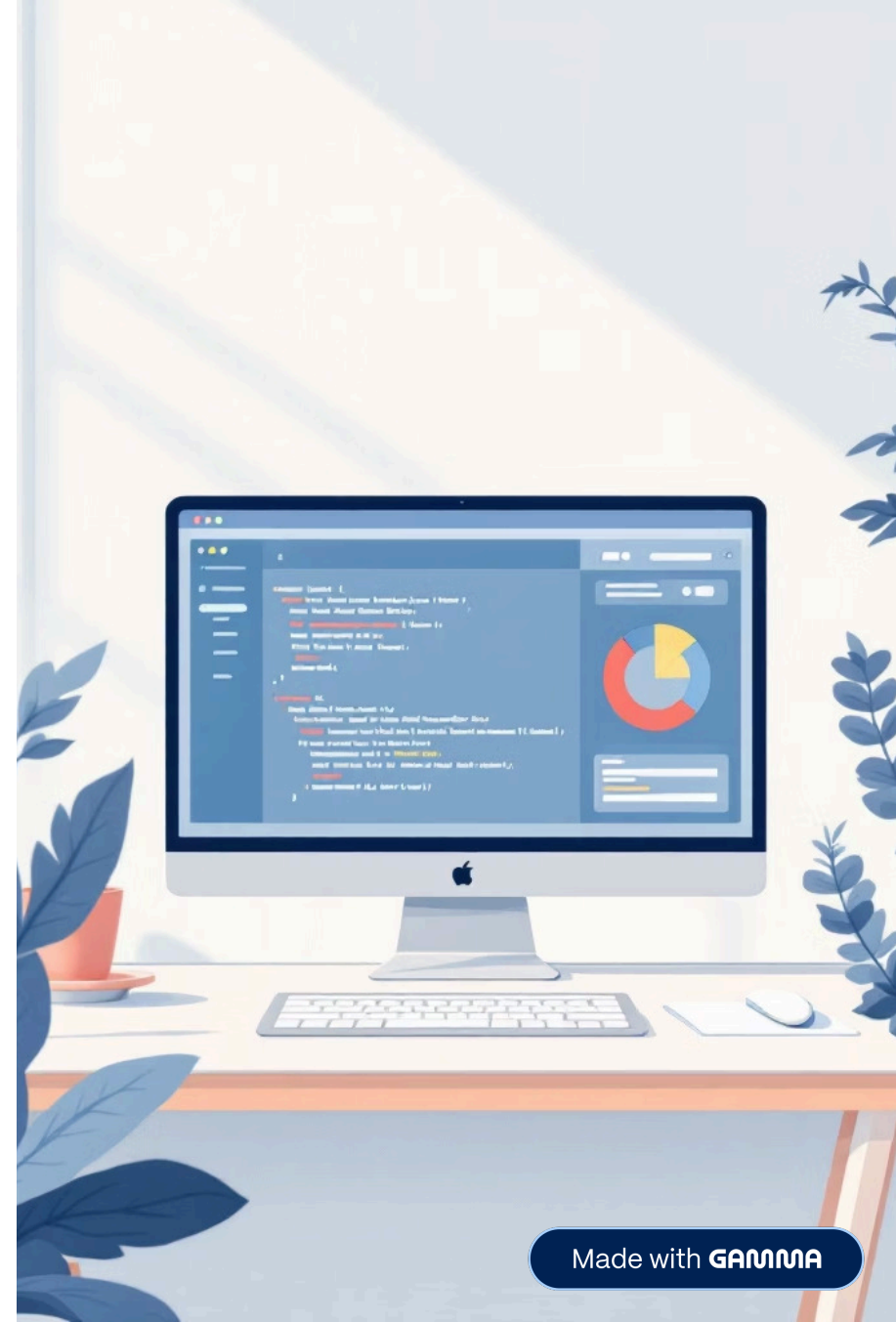
Shipping Comparison

Standard: \$58.46 | Express: \$60.48

4

Discount-Dependent Products

Hat (50%), Sneakers (49.66%), Coat (49.07%)



Customer Behavior Dashboard

Subscription Status

No Yes

Gender

Female Male

Category

- ☐ Accessories
- ☐ Clothing
- ☐ Footwear
- ☐ Outerwear

Shipping Type

- ☐ 2-Day Shipping
- ☐ Express
- ☐ Free Shipping
- ☐ Next Day Air
- ☐ Standard
- ☐ Store Pickup

4K

Total Customer

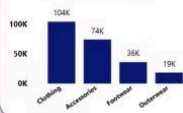
3.75

Average Review Rating

233.081K

Total Revenue (USD)

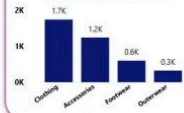
Revenue By Category



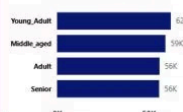
% of Customers by Subscription Status



Customers By Category



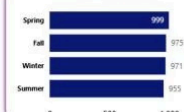
Revenue By Age Group



Customers by Payment Methods



Customers By Season



Power BI Dashboard

4K

Total Customers

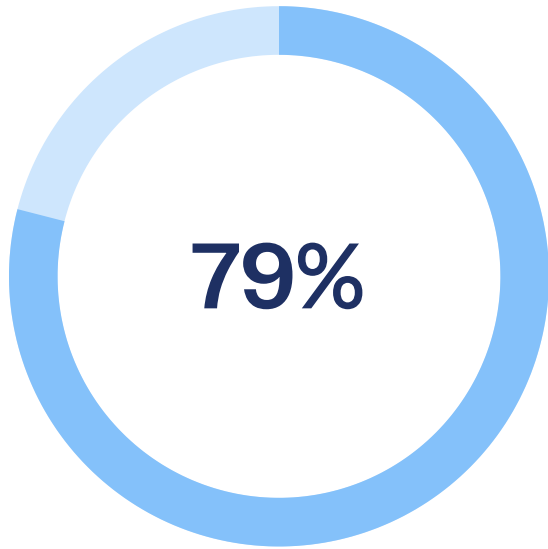
\$233K

Total Revenue

3.75

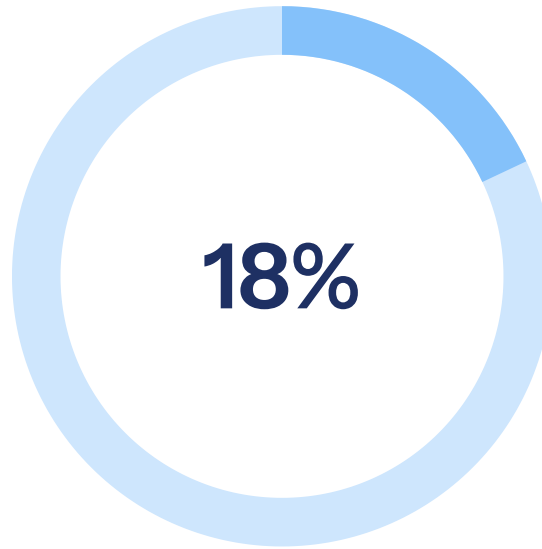
Average Rating

Customer Segmentation



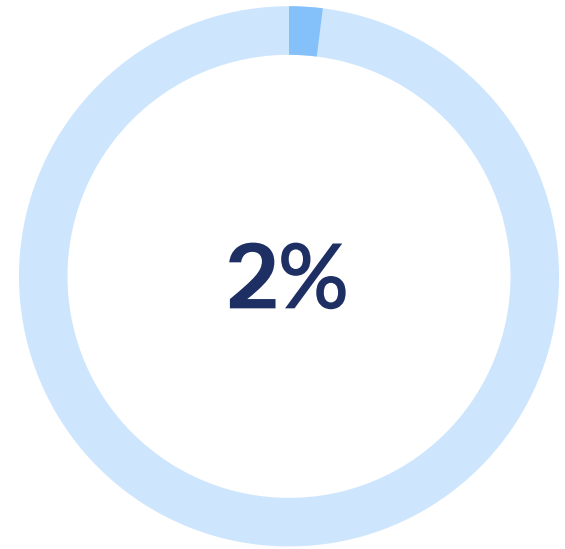
Loyal Customers

3,116 customers with consistent purchasing behavior



Returning Customers

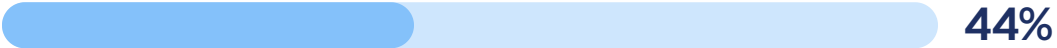
701 customers making repeat purchases



New Customers

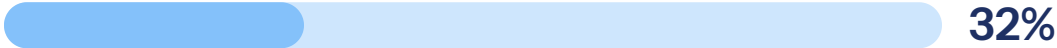
83 customers in initial purchase phase

Revenue Performance by Category



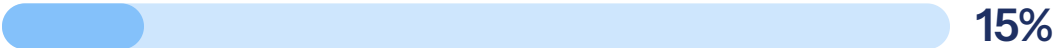
Clothing

\$104K revenue, 1.7K customers



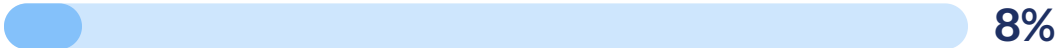
Accessories

\$74K revenue, 1.2K customers



Footwear

\$36K revenue, 600 customers



Outerwear

\$19K revenue, 300 customers

Key Insights



Clothing Dominates

Most purchased category with 1.7K customers and \$104K revenue



Male Customers Lead

Larger purchase portion, varied spending patterns across categories



Young Adults Drive Revenue

Young adults and middle-aged customers contribute highest revenue



Free Shipping Preferred

Standard and free shipping most frequently used delivery methods



Loyalty Strong

3,116 loyal customers indicate strong retention potential



Subscription Opportunity

Only 27% subscribed—significant growth potential

Business Recommendations



Boost Subscription Programs

Promote exclusive benefits and discounts for subscribed customers



Develop Customer Loyalty Programs

Introduce rewards for repeat customers to build loyal segment



Optimize Discount Strategies

Review policies to increase sales without affecting margins



Product Positioning

Highlight top-rated and best-selling products in campaigns



Targeted Marketing

Focus on high-revenue age groups and frequent shoppers